



A Radar and Sensor-Aware Kalman Filter Approach for Advanced Driver Assistance Systems

Chethana Chandrashekaraiah^{1*}, Deepu Raju²

¹ Department of Computer Science and Engineering, P.E.S. College of Engineering, Visvesvaraya Technological University, 590018 Mandya, India

² Department of Computer Science and Engineering, ATME College of Engineering, Visvesvaraya Technological University, 590018 Mysore, India

* Correspondence: Chethana Chandrashekaraiah (chethanac@pesce.ac.in)

Received: 01-23-2026

Revised: 04-13-2026

Accepted: 05-07-2026

Citation: C. Chandrashekaraiah and D. Raju, "A radar and sensor-aware Kalman filter approach for advanced driver assistance systems," *Int. J. Transp. Dev. Integr.*, vol. 10, no. 2, pp. 534–552, 2026. <https://doi.org/10.56578/ijtdi.100213>.



© 2026 by the author(s). Licensee Acadlore Publishing Services Limited, Hong Kong. This article can be downloaded for free, and reused and quoted with a citation of the original published version, under the CC BY 4.0 license.

Abstract: Advanced driver assistance systems (ADAS) rely heavily on robust object tracking to ensure safe and autonomous navigation, especially in complex outdoor environments. Traditional Kalman filter (KF)-based methods, while effective in ideal conditions, often fall short in scenarios with high noise, asynchronous sensor data, occlusions, and varying environmental conditions. The existing tracking techniques do not adequately address the challenges of multi-object tracking under low Signal-to-Noise Ratio (SNR) or nonlinear dynamics. To bridge this gap, this work proposes Radar and Sensor-Based Tracking with Adaptive Spatial-Temporal Analysis (RASTA), a modified KF-based architecture designed to enhance multi-object tracking using mmWave radar in ADAS. The primary objective of this work was to improve tracking accuracy, handle sensor uncertainty, and enable robust performance in dynamic and noisy conditions. The methodology involved simulating ADAS motion using a discrete Langevin process with bistable dynamics, converting Cartesian trajectories to polar coordinates, and introducing noise to emulate real-world radar behavior. Experimental validation using a mmWave dataset showed that RASTA achieved up to 12.4% improvement in azimuth estimation and 10.7% in radial distance accuracy over baseline methods. The results show RASTA's effectiveness in delivering reliable, accurate tracking.

Keywords: Advanced driver assistance systems; Kalman filter; mmWave radar; Multi-object tracking; Sensor fusion; Low Signal-to-Noise Ratio

1 Introduction

Vehicles have long served as the backbone of personal and commercial transportation, evolving from simple mechanical systems into complex, intelligent machines equipped with advanced computing, communication, and sensing capabilities. With the increasing demand for safety, efficiency, and autonomy, modern vehicles are now equipped with an array of electronic sensors and systems that facilitate functions ranging from navigation to collision avoidance [1, 2]. These advancements have made vehicles not just modes of transport, but intelligent entities capable of perceiving and reacting to their environments in real-time. Over the past two decades, significant progress has been made in integrating various sensors into vehicles to support tracking, object identification, and driver assistance functions [3].

Technologies such as Light Detection and Ranging (LiDAR), ultrasonic sensors, infrared cameras, Global Positioning Systems (GPS), Inertial Measurement Units (IMUs), and particularly mmWave radars are strategically placed around vehicles to ensure 360-degree perception [4, 5]. Among these, mmWave radar has gained significant attention due to its robustness under adverse weather and lighting conditions. However, despite these advantages, multi-object tracking using mmWave radar in low Signal-to-Noise Ratio (SNR) environments remains a critical challenge. Signal degradation due to rain, fog, multipath reflections, and dynamic occlusions introduces significant uncertainty, leading to inaccurate object localization and tracking instability [6].

In recent years, these sensing capabilities have been extended into Vehicular Ad-Hoc Networks (VANETs) and advanced driver assistance systems (ADAS) [7, 8], where Kalman filter (KF)-based approaches are widely used for

tracking and state estimation [9–20]. Although KF variants such as extended Kalman filter (EKF) and unscented Kalman filter (UKF) provide reasonable performance under moderate noise conditions, they exhibit limitations in handling nonlinear dynamics, state-dependent noise, and multi-object interactions in cluttered environments [21, 22]. Furthermore, sensor fusion introduces additional challenges due to asynchronous measurements, sensor drift, and varying reliability [10, 11].

Therefore, the key technical problem addressed in this work is the development of a robust multi-object tracking framework capable of operating under low-SNR, nonlinear, and cluttered ADAS environments using mmWave radar data. To address this, we propose Radar and Sensor-Based Tracking with Adaptive Spatial-Temporal Analysis (RASTA), a modified KF-based framework that integrates adaptive noise modeling, nonlinear dynamics, and spatial-temporal filtering to enhance tracking accuracy and robustness.

2 Literature Survey

Recent advancements in intelligent transportation systems (ITS) and ADAS have significantly improved object detection and tracking capabilities. Existing research in this domain can be broadly categorized into deep learning (DL)-based detection methods, KF-based tracking approaches, and sensor fusion techniques, each contributing uniquely to the development of robust vehicle perception systems.

2.1 Deep Learning-Based Detection Approaches

DL models have shown remarkable performance in object detection tasks within autonomous driving environments. Ren et al. [12] proposed an enhanced YOLOv8-based framework incorporating dynamic-head structures and coordinate-attention mechanisms to improve detection accuracy. Their approach effectively addressed feature representation and scaling challenges, achieving notable performance gains on standard datasets. However, such DL-based methods primarily focus on detection and lack temporal consistency, making them less suitable for continuous tracking in dynamic environments. Additionally, they are sensitive to noise, occlusion, and adverse weather conditions, particularly when relying solely on visual data.

2.2 Kalman Filter-Based Tracking Approaches

KF and its variants remain fundamental tools for state estimation and tracking in ADAS systems due to their computational efficiency and recursive nature. Negru et al. [13] employed an EKF within a federated fusion framework for UAV navigation, integrating multiple sensors such as IMU, GNSS, and vision data. While EKF improves nonlinear estimation through linearization, it suffers from approximation errors under highly nonlinear conditions. Similarly, Liu et al. [14] utilized a UKF for radar-camera fusion in traffic scenarios, demonstrating improved tracking accuracy through nonlinear transformation techniques. However, UKF introduces higher computational complexity and may become unstable under severe noise conditions. Cao et al. [16] proposed a robust UKF-based approach combined with optimization techniques for underground positioning, achieving improved accuracy but lacking applicability in real-time multi-target vehicular environments. Yang et al. [20] introduced an adaptive extended Kalman filter (AEKF) for beam prediction in mmWave communication systems, dynamically adjusting noise covariance to improve estimation accuracy. Despite these advancements, most KF-based approaches assume Gaussian noise and struggle with state-dependent uncertainties, cluttered environments, and multi-object interactions. Recent developments, such as multi-stage Kalman filtering systems [21], have improved sensor fusion efficiency; however, they still rely on simplified assumptions that limit performance in complex ADAS scenarios.

2.3 Sensor Fusion and Radar-Based Tracking Approaches

Sensor fusion techniques aim to combine complementary information from multiple sensors to enhance robustness. Moosavi et al. [15] proposed a Sensor-Health-Aware Resilient Fusion (SHARF) framework, which dynamically adjusts sensor weights based on reliability. While effective in handling faulty sensors, it does not explicitly address occlusion and asynchronous data challenges. Sarkar et al. [17] introduced CarVision, a radar-based vehicle tracking system that integrates DL with mmWave radar for distance estimation. Although effective in controlled conditions, its performance is limited under high noise and clutter. Zhu et al. [18] proposed DeepEgo+, which handles unsynchronized radar inputs using decentralized neural networks; however, it lacks robust filtering mechanisms for handling measurement uncertainty. Venkatesha et al. [19] developed Multivariate Fuzzy Density-Based Temporal-Spatial Clustering of Applications with Noise (MFDBTSCAN), a clustering-based approach for radar target association, improving detection accuracy in multi-target scenarios. However, it does not fully address nonlinear motion dynamics or noise propagation. Recent studies [22–25] further highlight the challenges associated with mmWave radar, including low SNR, multipath interference, and environmental variability, which significantly affect tracking performance in real-world ADAS deployments.

2.4 Research Gap

Despite extensive research, several limitations persist as presented in Table 1. DL approaches lack robustness to noise and temporal consistency, while KF-based methods struggle with nonlinear dynamics and state-dependent noise. Sensor fusion techniques, although effective, face challenges related to asynchronous data and reliability. Moreover, existing radar-based methods are highly sensitive to clutter, occlusion, and low SNR conditions. To address these challenges, there is a need for a robust, adaptive tracking framework capable of handling nonlinear system behavior, dynamic noise conditions, and multi-object interactions in real-world ADAS environments. The proposed RASTA framework aims to fill this gap by integrating adaptive noise modeling, nonlinear state estimation, and spatial-temporal filtering within a modified Kalman filtering structure.

Table 1. Comparative summary of existing tracking approaches

Ref.	Approach Type	Application Scenario	KF Variant	Multi-Target	Noise Handling	mmWave Radar	Key Limitation
[12]	DL detection	Autonomous driving	No	Yes	Low	No	No filtering, noise sensitivity
[13]	Sensor fusion	UAV navigation	EKF	No	Moderate	No	Linearization errors
[14]	Radar-vision fusion	Traffic monitoring	UKF	Yes	High	Yes	High complexity
[15]	Sensor fusion	Multi-vehicle tracking	KF	Yes	High	Yes	No occlusion handling
[16]	Localization	Underground	UKF	No	High	No	No real-time multi-target
[17]	Radar tracking	Vehicle ranging	No	No	Moderate	Yes	Limited robustness
[18]	Radar fusion	Ego-motion	No	No	Moderate	Yes	No filtering robustness
[19]	Clustering + KF	Radar tracking	KF	Yes	High	Yes	No nonlinear modeling
[20]	Beam prediction	V2I	AEKF	No	Moderate	Yes	Limited clutter handling
[21]	Multi-stage KF fusion	Sensor fusion systems	Multistage KF	Yes	High	No	Assumes Gaussian noise, limited nonlinear handling
[22]	Collision avoidance	ADAS (foggy conditions)	No	Yes	High	Yes	Focus on avoidance, not tracking accuracy
RASTA [proposed]	Adaptive KF framework	ADAS multi-object tracking	DNKF (modified KF)	Yes	Very high (adaptive + statedependent)	Yes	Addresses nonlinear dynamics, clutter, occlusion, and low-SNR

Note: Deep learning (DL), Kalman filter (KF), extended Kalman filter (EKF), unscented Kalman filter (UKF), adaptive extended Kalman filter (AEKF), Autonomous Driving Assistant System (ADAS), dynamic non-linear Kalman filter (DNKF), Signal-to-Noise-Ratio (SNR), unmanned aerial vehicle (UAV), and Vehicle-to-Infrastructure (V2I).

3 Methodology

This section begins by presenting the overall architecture of the proposed RASTA system, followed by its underlying modeling framework. First, the conventional KF is introduced along with its limitations in complex ADAS environments. Subsequently, the proposed modifications are described in detail to enhance the KF for

nonlinear and noisy conditions. Finally, the prediction and correction procedures within the modified KF structure are presented, highlighting how these adaptations improve tracking accuracy and robustness.

To provide a clear structural understanding, the proposed RASTA (Robust Adaptive State Tracking Architecture) is a comprehensive ADAS-oriented tracking framework designed to operate effectively in cluttered, dynamic, and uncertain environments. The framework is composed of three primary layers:

1. Sensing Layer: This layer consists of mmWave radar and onboard sensors that continuously collect environmental data, including object position, velocity, and Doppler information.
2. Filtering Layer (DNKF Core): At the core of the RASTA framework lies the proposed Dynamic Nonlinear Kalman filter (DNKF), which serves as the primary estimation engine. It performs state prediction and correction by incorporating nonlinear system dynamics and adaptive noise modeling to handle uncertainties, clutter, and occlusions.
3. Decision Layer: The refined state estimates generated by the DNKF are forwarded to the ADAS processing unit, which utilizes this information for path planning, collision avoidance, and driver assistance functions.

It is important to note that RASTA represents the overall system-level framework, while DNKF constitutes its core filtering component. Specifically, DNKF acts as the mathematical backbone of RASTA by extending the traditional KF to support nonlinear modeling and state-dependent noise adaptation. This integration enables robust and accurate state estimation in real-world ADAS scenarios characterized by dynamic motion, sensor noise, and environmental uncertainties.

3.1 Architecture

For understanding the architecture of this work, in Figure 1, an ADAS environment encountered during real-world scenarios is presented. In Figure 1, the environment consists of multiple vehicles, where some vehicles have mmWave radars/sensors and some vehicles do not have mmWave radars/sensors. Also, Figure 1 shows the static objects on a lane, like road dividers and trees, along with dynamic environment elements like curved roads and occlusions. These factors contribute to signal interference and tracking difficulty. Hence, capturing the essence of a cluttered and noisy environment for robust multi-object tracking is important for ensuring vehicle safety and performance.

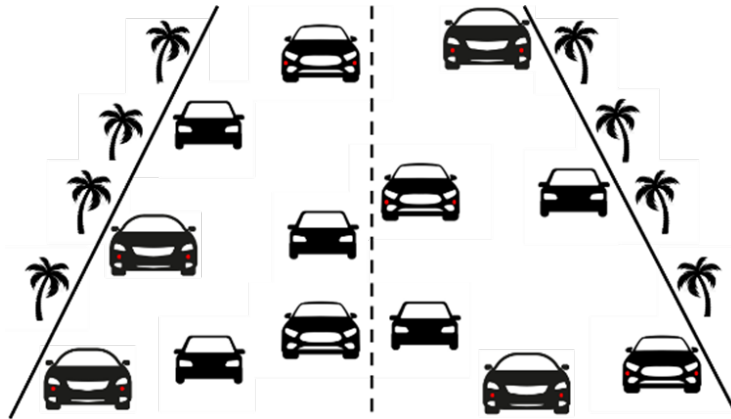


Figure 1. Example of an Advanced Driver Assistance Systems (ADAS) environment

For solving the issue of cluttered and noisy environments for robust multi-object tracking, this work presents the RASTA architecture presented in Figure 2.

The process begins with mmWave radars and sensors mounted on the vehicle, typically positioned on the front bumper, sides, and rear, which continuously collect spatial and motion data of surrounding dynamic and static objects. This sensor data, including position, velocity, and Doppler shifts, is then input into the RASTA module. RASTA operates in two key stages: prediction and correction, initialized from a known or estimated initial state. During the prediction phase, the current state is projected forward in time using a dynamic motion model, and the upcoming state is estimated, accounting for potential nonlinearities and uncertainties. In the correction phase, the incoming radar observations are used to compute the RASTA gain (a modified Kalman gain), which is then used to update the system state and correct the predicted estimate. Subsequently, the error covariance matrix is updated to reflect the uncertainty of the corrected estimate, ensuring robustness in environments with clutter or sensor noise. The output of the RASTA module, accurate, real-time estimates of surrounding object states, is passed to the ADAS Processing Unit, which uses this refined information for path planning, driver alerts, and object avoidance maneuvers. Feedback loops within RASTA (as indicated by the orange arrows) ensure continuous refinement of predictions based on prior corrections, maintaining high tracking accuracy in highly dynamic and occluded scenes. The complete model of RASTA is discussed in detail in the next section.

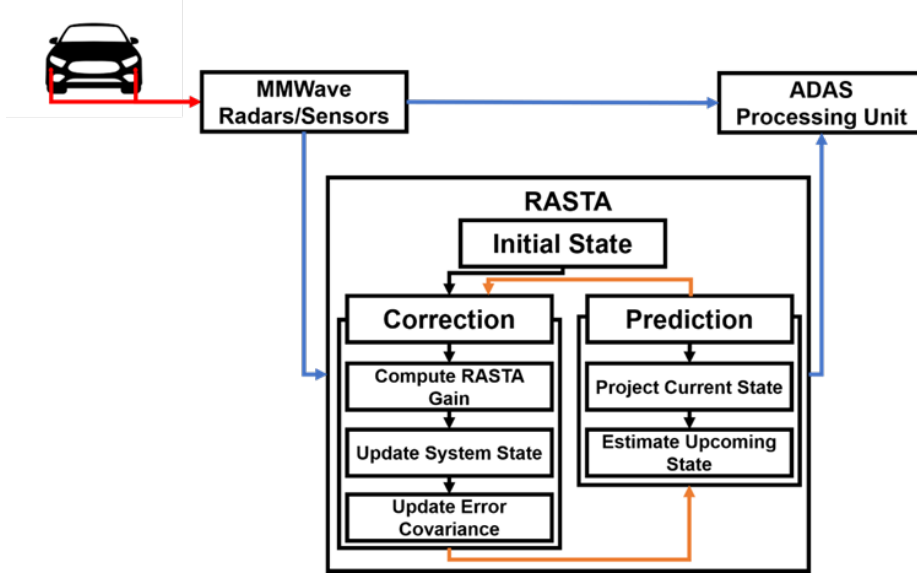


Figure 2. RASTA architecture

Note: Radar and Sensor-Based Tracking with Adaptive Spatial-Temporal Analysis (RASTA); Advanced Driver Assistance Systems (ADAS).

3.2 Radar and Sensor-Based Tracking with Adaptive Spatial-Temporal Analysis

For understanding how RASTA contributes to object tracking in ADAS environments, this work first discusses the standard KF approach. This section lays the mathematical foundation before presenting proposed modifications for handling real-world noise and uncertainty. The details of all notation to derive the RASTA are provided in Table 2.

3.2.1 Traditional Kalman filter

In a stochastic environment, the system dynamics are represented using state transition and measurement, which are represented in Eq. (1) and Eq. (2). Eq. (1) represents the state transition model, where the future state depends on the current state and process noise.

$$y_{l+1} = B_l y_l + \delta_l x_l \quad (1)$$

In Eq. (1), $y_i \in \mathbb{S}^o$ which denotes state-vector at time-step l , which contains parameters such as position, velocity orientation of vehicle, $B_l \in \mathbb{S}^{o \times o}$ which denotes know state transition matrix describing how vehicle state evolves from step l to $l + 1$, $\delta_l \in \mathbb{S}^{o \times n}$ denotes process noise influence matrix, representing impact of Gaussian noise x_l on the state, x_l denotes zero-mean Gaussian process noise with know optimized covariance. The x_l represents uncertainties in the system (e.g., due to wind or actuator noise). Eq. (2) represents the measurement model, where sensor observations are obtained as a noisy projection of the system state.

$$z_l = I_l y_l + w_l \alpha_l \quad (2)$$

Further, in Eq. (2), $z_l \in \mathbb{S}^q$ denotes measurement vector collected from sensors, $I_l \in \mathbb{S}^{q \times o}$ denotes measurement matrix, mapping state vector into measurement space, $w_l \in \mathbb{S}^{q \times o}$ denotes measurement noise influence matrix, determining how noise affects observations, α_l denotes zero-mean, Gaussian measurement noise, which is uncorrelated with x_l , $S_l = w_l w_l^T$ denotes measurement noise covariance matrix, quantifying measurement uncertainty, $R_l = \delta_l \delta_l^T$ denotes process noise covariance matrix, representing uncertainty in model dynamics. In the KF update phase, the KF is used to recursively estimate the state y_i using noisy observations. The update process involves inverse prior covariance $H \leftarrow Q_{l|l-1}$, inverse measurement covariance $X \leftarrow S_l^{-1}$, prediction error $w \leftarrow \hat{y}_{l|l} - \hat{y}_{l|l-1}$, observation matrix $G \leftarrow I_l$, and innovation (measurement residual) $\alpha \leftarrow z_l - I_l \hat{y}_{l|l-1}$. In the update process, $\hat{y}_{l|l-1}$ is the predicted state estimate from the previous time step, which is evaluated as $\hat{y}_{l|l-1} = B_{l-1} \hat{y}_{l-1|l-1}$, and $Q_{l|l-1}$ is the predicted error covariance matrix, which is evaluated as $Q_{l|l-1} = B_{l-1} Q_{l-1|l-1} B_{l-1}^T + R_{l-1}$. For solving the estimation problem, the KF uses the Regularized Least Squares (RLS) formulation. Consider that an estimate is required for an unknown vector $y \in \mathbb{S}^o$ from measurements $z \in \mathbb{S}^q$, then $z = Gy + \alpha$, $G \in \mathbb{S}^{q \times o}$, denotes the observation matrix, and α represents the measurement noise. Consider $\bar{y}$, which denotes the prior estimate of y , $w = y - \bar{y}$ denotes the deviation from the prior, and $\alpha = z - G\bar{y}$ denotes the measurement innovation. Then, the RLS cost function is evaluated using Eq. (3).

$$\min_w [w' H_w + (G_{w-\alpha})' X (G_w - \alpha)] \quad (3)$$

Table 2. Notation table and description

Symbol	Description
y_l	State vector at time step l (position, velocity, etc.)
z_l	Measurement vector from radar/sensors
B_l	State transition matrix
I_l	Measurement matrix
δ_l	Process noise influence matrix
w_l	Measurement noise influence matrix
x_l	Process noise (Gaussian, zero mean)
α_l	Measurement noise (Gaussian, zero mean)
Q_l	Error covariance matrix
R_l	Process noise covariance
S_l	Measurement noise covariance
L_l	Kalman gain
$g(\cdot)$	Nonlinear state transition function
$i(\cdot)$	Nonlinear measurement function
$\tilde{y}_l$	Deviation from nominal state
$D(\cdot)$	State-dependent nonlinear scaling function
δ_y, δ_y^-	State noise modeling coefficients
δ_z, δ_z^-	Measurement noise modeling coefficients
α_{ly}	Additional Gaussian noise terms
w_*	Optimal correction term
γ_l	Residual correction term
M_l	Noise adaptation gain
J	Covariance of additional noise
$F_l \cdot$	Prediction operator
$Q_{l l}$	Posterior error covariance matrices
$Q_{l+1 l}$	Prior error covariance matrices

In Eq. (3), $H = H' \succ 0$, which denotes a positive definite prior weight matrix, encoding confidence in the prior, $X = X' \succ 0$, which denotes a positive definite measurement weight matrix, representing trust in sensor accuracy, and $G \in \mathbb{S}^{q \times o}$ denotes the observation matrix (e.g., I_l). Eq. (3) defines the regularized least squares objective, which minimizes both deviation from prior estimate (model confidence) and measurement error (sensor confidence). This formulation provides the optimization foundation for the KF update. For achieving the optimal solution, Eq. (3) is solved, which provides Eq. (4).

$$w^* = [H + G' X G]^{-1} G' X_\alpha \quad (4)$$

Eq. (4) gives the optimal correction term by balancing prior uncertainty and measurement reliability using weighted matrices H and X . Using the optimal solution from the RLS approach, the final KF update equations for state update, Kalman gain, and updated error covariance are obtained using Eq. (5), Eq. (6), and Eq. (7), respectively.

$$\hat{y}_{l|l} = \hat{y}_{l|l-1} + L_l (z_l - I_l \hat{y}_{l|l-1}) \quad (5)$$

$$L_l = Q_{l|l-1} I_l' S_l^{-1} \quad (6)$$

$$Q_{i|l} = \left(Q_{l|l-1}^{-1} + I_l' S_l^{-1} I_l \right)^{-1} \quad (7)$$

Using Eq. (5), Eq. (6), and Eq. (7), the KF performs real-time correction of predicted states based on incoming observations, while the Regularized Least Squares (RLS) interpretation reveals the underlying optimization

mechanism that balances confidence between model predictions and sensor measurements. Specifically, Eq. (5) defines the state update, where the predicted state is corrected using the measurement residual; Eq. (6) represents the Kalman gain, which determines the extent to which the measurement influences the update; and Eq. (7) corresponds to the covariance update, which adjusts the uncertainty associated with the estimated state after correction. Although conventional KF provides reliable performance under linear and Gaussian assumptions, it often degrades in cluttered and noisy environments, particularly in robust multi-object tracking scenarios. Therefore, in this work, the KF is modified to address these limitations and enhance performance under real-world conditions.

3.2.2 Modified Kalman filter

In a typical ADAS tracking system, KF approaches have been widely adopted due to their efficiency in linear and Gaussian environments, as discussed in the literature survey. However, existing KFs tend to degrade performance in a cluttered and noisy environment for robust multi-object tracking or in scenarios where prior state information is incomplete or unavailable. Hence, this work presents a modified KF, namely DNKF (dynamic non-linear Kalman filter), suitable for non-linear environments, incorporating state and measurement linearization with noise approximations. In the modified KF, the state evolution and measurement are evaluated using Eq. (8) and Eq. (9), respectively.

$$y_{l+1} = g(y_l) + \delta x_l \quad (8)$$

$$z_l = i(y_l) + w\alpha_l \quad (9)$$

These equations extend the KF to nonlinear dynamics, where, $g(\cdot)$ models nonlinear state evolution and $i(\cdot)$ models nonlinear measurement mapping. In Eq. (8), $y_l \in \mathbb{R}^o$, which denotes the state vector at time l (velocity, position), $g(y_l)$ denotes a non-linear function representing the system's dynamics, and δx_l denotes a noise term, where, δ is the process noise coefficient matrix, and x_l denotes Gaussian noise with zero mean. In Eq. (9), $z_l \in \mathbb{R}^q$, which denotes the measurement vector obtained from sensors, $i(y_l)$ denotes the non-linear measurement function, and $w\alpha_l$ denotes the measurement noise term with w as the noise influence matrix and α_l denotes zero-mean Gaussian noise.

For enabling filtering in a non-linear context, this work has applied linearization around the equilibrium point $y_l = \bar{y}$. This approximation leads to Eq. (10).

$$y_{l+1} = g(\bar{y}) + \left. \frac{\partial g}{\partial y} \right|_{y=\bar{y}} (y_l - \bar{y}) + P \left(\|y_l - \bar{y}\|^2 \right) + \delta x_l \quad (10)$$

Eq. (10) applies a first-order Taylor expansion to approximate nonlinear dynamics around a nominal point. The higher-order term captures residual nonlinear effects. In Eq. (10), $\bar{y}$ denotes the chosen equilibrium or nominal point for linearization, $\left. \frac{\partial g}{\partial y} \right|_{y=\bar{y}}$ denotes the Jacobian matrix of the non-linear function g_2 , evaluated at $\bar{y}$, representing linearized dynamics, $P(\|y_l - \bar{y}\|^2)$ denotes a higher-order non-linear term, which captures residual modeling error due to linearization. Consider $\tilde{y}_l = y_l - \bar{y}$, which denotes deviation from equilibrium points, $B = \left. \frac{\partial g}{\partial y} \right|_{y=\bar{y}}$ which denotes the linearized state transition matrix and $\sigma g_l = g(\bar{y})$, which denotes the constant part of the non-linear system evaluated at equilibrium. From this, the KF can be reformulated as Eq. (11).

$$\tilde{y}_{l+1} = B\tilde{y}_l + \sigma g_l + P \left(\|\tilde{y}_l\|^2 \right) + \delta x_l \quad (11)$$

The Eq. (11) reformulates the system into a KF-compatible structure, separating linear dynamics and nonlinear residual terms. Further, for handling performance in a cluttered and noisy environment for robust multi-object tracking, the higher-order term $\sigma g_l = P \left(\|\tilde{y}_l\|^2 \right)$ is approximated using the error structure defined in Eq. (12).

$$\sigma g_l \cong \left(\delta_y + \tilde{\delta}_y D(\|\tilde{y}_l\|) \right) \alpha_l^y \quad (12)$$

Eq. (12) models state-dependent noise, where noise magnitude varies based on estimation error, using a function $D(\cdot)$. In Eq. (12), $\delta_y, \tilde{\delta}_y$ denote coefficient matrices used for modeling non-linear approximation noise, $D(\cdot)$ denotes a non-linear function that adjusts the effect based on the magnitude of deviation, α_l^y denotes zero-mean Gaussian noise component linked to state error. By incorporating the above noise approximation, the updated enhanced state and enhanced measurement model become as presented in Eq. (13) and Eq. (14).

$$y_{l+1} = B y_l + \delta x_l + \left(\delta_y + \tilde{\delta}_y D(\|\tilde{y}_l\|) \right) \alpha_l^y \quad (13)$$

$$z_l = I y_l + w \alpha_l + (\delta_z + \bar{\delta}_z D(|\tilde{y}_l|)) \alpha_l^z \quad (14)$$

These equations incorporate adaptive noise terms into state equations, i.e., Eq. (13), and measurement equations, i.e., Eq. (14), enabling robustness in cluttered and uncertain environments. In Eq. (13) and Eq. (14), I denotes the measurement matrix, δ_z , $\bar{\delta}_z$ denotes coefficients representing measurement noise modeling matrices, α_l^z denotes a Gaussian noise term for the measurement side, B denotes the state transition matrix derived from linearization, and $|\tilde{y}_l|$ denotes the error in the state estimate from the nominal value. In this modified KF, the random variables involved in system x_l , α_l^y , α_l , α_l^z are all assumed to follow zero-mean Gaussian distributions with uniform covariance matrices. The matrices include $\delta \delta^U \succ 0$, which defines process noise covariance and $w w^U \succ 0$, which defines measurement noise covariance. These conditions ensure positive definiteness, which is important for KF convergence and stability. In this work, for each time step l , the modified KF performs two major operations, which include prediction and correction. In prediction, the modified KF estimates the state and its covariance based on previous data and the dynamic model, and in the correction, the predictions are refined using new measurements z_l . Consider $\hat{y}_{l|l}$ as the updated state estimate at step l , $Q_{l|l}$ as the associated error covariance, and z_l as the measurement at time l . Then the modified KF computes the expected path using Eq. (15).

$$F\{z_l, \hat{y}_{l|l}, Q_{l|l} \mid l\} = F\{ \cdot \} \quad (15)$$

This modified KF approach enables effective state estimation in non-linear tracking environments. By accounting for higher-order system behavior, adaptively modeling noise, and avoiding dependence on prior information, this method enhances robustness and tracking accuracy under real-world conditions. In the next section, the modified KF prediction and correction process is discussed in detail.

3.3 Prediction and Correction Process in Modified Kalman Filter for Radar and Sensor-Based Tracking with Adaptive Spatial-Temporal Analysis

For prediction, this work considers an initial condition for the system state as $y_0 \sim O(\hat{y}_0, Q_0)$. This means the initial state y_0 is Gaussian-distributed with known prior mean $\hat{y}_0$ and known covariance Q_0 . Given that the state noise x_l and modeling noise α_{ly} are statistically independent, the predicted state estimate at time $l+1$, denotes as $\hat{y}_{l+1|l} = F_l\{y_{l+1}\}$ becomes $\hat{y}_{l+1|l} = B \hat{y}_{l|l}$, where $\hat{y}_{l+1|l}$ is predicted state at time $l+1$ based on data up to time l , B denotes linearized system matrix derived from non-linear dynamics and $\hat{y}_{l|l}$ denotes updated state estimate at time l . However, because the system involves state-dependent noise, estimating the error covariance matrix becomes more complex. Hence, this work defines a state-dependent noise adjustment function as $Y_l = Y_l\{D(|y_l|)\}$, where the $D(\cdot)$ captures dependence of noise magnitude on the size of the state vector. For the modified state presented in Eq. (13), the predicted covariance of next state changes is presented in Eq. (16).

$$Q_{l+1|l} = B Q_{l|l} B^U + R_l \quad (16)$$

Eq. (16) updates the predicted uncertainty, combining propagated previous uncertainty and adaptive process noise R_l . In Eq. (16), $Q_{l+1|l}$ denotes the predicted covariance matrix at time $l+1$, $Q_{l|l}$ denotes the updated covariance at time l , and R_l denotes the total process noise covariance matrix at time l , which includes both standard and state-dependent components. The R_l is evaluated as $R_l = \delta \delta^U + \delta_y \delta_y^U + \bar{\delta}_y D(Q_{l|l} + \hat{y}_{l|l} + \hat{y}_{l|l} \hat{y}_{l|l}^U) \bar{\delta}_y^U$. Once the new measurement z_{l+1} is received, the KF correction step begins. In traditional KF, the state update is obtained by minimizing the cost function defined in Eq. (17).

$$\min_y \left(\|y - \hat{y}_{l+1|l}\|^2 + \|z_{l+1} - I y\|_{S^{-1}}^2 \right) \quad (17)$$

This equation defines the optimization problem for correction, minimizing the prediction error and the measurement residual using the Mahalanobis distance. In Eq. (17), I denotes the measurement matrix, $S = w w^U \succ 0$, which denotes the measurement noise covariance, and $\|\cdot\|_{S^{-1}}^2$ denotes the Mahalanobis distance using the inverse of the noise covariance matrix. For ensuring modified RF remains well-defined, this work has enforced $R_l \succ 0$. Hence, using this, the updated state estimate is updated as $\hat{y}_{l+1|l+1} = \hat{y}_{l+1|l} + w^*$, where w^* is a correction term that minimizes the combined uncertainty in both prediction and measurement, which is evaluated using Eq. (18).

$$w^* = \arg \min_w \left(\|w\|_{Q_{i+1}^{-1} l+1}^2 + \|a_{l+1} - w\|_{S^{-1}}^2 \right) \quad (18)$$

Eq. (18) computes the optimal deviation (correction term) between the predicted and true state. In Eq. (18), $w = y - \hat{y}_{l+1|i}$, which is estimation deviation from predicted values, and $a_{l+1} = y_{l+1} - I \hat{y}_{l+1|i}$ is residual between true

value and predicted measurement. For enhancing accuracy, uncertainty modeling is included in the minimization function, defined in Eq. (19).

$$w^* = \arg \min_w F \left\{ \|w\|_{Q_{i+1/i+1}^{2-1}}^2 + \|a_{l+1} - I_w + (\delta_w + \bar{\delta}_w D(|w|)) \alpha_{l+1}^w\|_{S^{-1}}^2 \right\} \quad (19)$$

This extends Eq. (18) by including state-dependent and adaptive noise, improving robustness under uncertainty. In Eq. (19), $\delta_w, \bar{\delta}_w$ denote new adaptive noise matrices modeling uncertainty in measurement and $\alpha_{l+1}^w \sim N(0, J)$ denotes additional zero-mean Gaussian noise with covariance J . With all corrections accounted for, the improved state update becomes as presented in Eq. (20).

$$\hat{y}_{l+1|l+1} = \hat{y}_{l+1} + \underbrace{L_{l+1} (z_{l+1} - \hat{y}_{l+1|l}) - M_{l+1} \gamma_{l+1}}_{w^*} \quad (20)$$

Eq. (20) provides the enhanced state update, combining standard Kalman correction and an additional noise compensation term. In Eq. (20), $L_{l+1} = Q_{l+1|l} I^U S^{-1}$, which denotes modified KF gain, which optimally weighs the innovation, $M_{l+1} = [J - \omega + \omega N_l = 1]^{-1} \omega$, which denotes correction gain for noise. $N_{l+1} = Q_{l+1|l}^{-1} + \Lambda (S^{-1})$ which denotes the inverse of the combined uncertainty and $\gamma_{l+1} = \frac{1}{2} \Delta (S^{-1}) T(w^*)$ which denotes residual correction. This modified KF method improves tracking performance in environments where system dynamics are non-linear, and noise is both state-dependent and uncertain. Further, to ensure consistency and stability of the covariance matrix in the modified KF, the error covariance matrix must fulfill the constraint given in Eq. (21).

$$\begin{aligned} & Q_{l+1|l} - L_{l+1} [\bar{\delta}_z D(Q_{l+1|l+1}) \bar{\delta}_z^U + \delta_z Y_{l+1} \bar{\delta}_z + \bar{\delta}_z Y_{l+1} \delta_z] L_{l+1}^U \\ &= (J - L_{l+1} I) Q_{l+1|l} (J - L_{l+1} I)^U + L_{l+1} \tilde{S}_{l+1} L_{l+1}^U + \beta_{l+1} \mu_{l+1} \beta_{l+1}^U \end{aligned} \quad (21)$$

This equation ensures stability and positive definiteness of the covariance matrix by incorporating adaptive noise, residual correction, and noise propagation effects. In Eq. (21), $Q_{l+1|l}$ denotes predicted state covariance matrix at time step $l+1$ based on information available up to time l , L_{l+1} denotes modified KF gain matrix at time step $l+1$, which determines how much prediction should be corrected by measurement, $\delta_z, \bar{\delta}_z$ denotes noise matrices associated with state-dependent and dynamic uncertainties in measurement process, $D(Q)$ denotes nonlinear scaling of uncertainty based on predicted covariance matrix, Y_{l+1} denotes state-dependent function used for incorporating dynamic uncertainty in error-term, J denotes transformation matrix used for residual correction, $\tilde{S}_{l+1}$ denotes extended measurement noise covariance matrix which is evaluated as $\tilde{S}_{l+1} = w w^U + \delta_z \delta_z^U + \bar{\delta}_z (\hat{y}_{l+1|l+1} \hat{y}_{l+1|l+1}^U) \bar{\delta}_z^U$, β_{l+1} denotes noise propagation correction term evaluated as $\beta_{l+1} = [M_{l+1} (J - L_{l+1} I)]$ and μ_{l+1} denotes uncertainty correction matrix defined using propagated and residual noise evaluated as $\mu_{l+1} = \begin{bmatrix} \gamma_{l+1} \gamma_{l+1}^U & \gamma_{l+1} n_{l+1}^U \\ n_{l+1}^U \gamma_{l+1} & 0 \end{bmatrix}$, where $n_{l+1} = B((J - L_l I) n_l + M_l \gamma_l)$. The integration of adaptive noise modeling via matrices $\delta_y, \delta_w, \bar{\delta}_y, \bar{\delta}_w$, and the inclusion of additional correction terms, enables more robust and accurate state estimation for ADAS in dynamic and unpredictable conditions. The complete process of the RASTA is presented in Algorithm 1. discussed below.

The RASTA framework presented in Algorithm 1 builds upon the conventional KF by introducing enhancements that improve robustness against occlusions, measurement clutter, and nonlinear process noise. It operates in an iterative manner, where at each time step, radar measurements are processed through the DNKF core to produce accurate state estimates. Specifically, the operation at each time step can be understood as a sequence of key steps. Initially (Step 1: Data Acquisition), mmWave radar sensors acquire measurements such as object position, velocity, and Doppler information from the surrounding environment. Using these observations, the DNKF performs a prediction step (Step 2: Prediction), where the future state of each object is estimated based on the previous state and system dynamics. Unlike the traditional KF, this prediction accounts for nonlinear motion behavior and state-dependent noise, enabling improved performance in complex scenarios such as curved roads and partial occlusions.

Following the prediction, the framework incorporates an adaptive noise modeling mechanism (Step 3: Noise Adaptation), where the noise characteristics are dynamically adjusted based on the magnitude of the estimation error. This allows the system to remain robust in highly cluttered and uncertain environments. In the correction step (Step 4: Correction), incoming radar measurements are compared with predicted values to compute the innovation (residual), which quantifies the discrepancy between expected and observed measurements. A modified Kalman gain is then computed to determine the extent to which the prediction should be corrected.

Subsequently, the state update step (Step 5: State Update) refines the predicted state by integrating both measurement information and adaptive noise compensation terms, resulting in a more accurate estimate. The associated error covariance is also updated (Step 6: Covariance Update) to reflect the revised uncertainty, incorporating both standard and nonlinear noise effects. Finally, a feedback mechanism (Step 7: Feedback Loop) ensures that the

updated state and covariance are propagated to the next iteration, enabling continuous refinement of the tracking process.

The entire procedure is mathematically formalized through Eq. (10) to Eq. (21), which collectively capture linearization, adaptive noise modeling, and enhanced correction strategies. This structured approach enables RASTA to effectively handle real-world ADAS scenarios, maintaining high tracking accuracy even under non-Gaussian noise conditions and imperfect measurements. The performance evaluation of the proposed RASTA framework is presented in the subsequent section.

Algorithm 1 RASTA

Inputs:

- Initial state estimate $\hat{y}_0$, initial covariance Q_0 .
- State and measurement noise models.
- Non-linear system functions $g(\cdot)$ and $i(\cdot)$.
- Radar measurements $(z_1, z_2, \dots, z_T)$.

Output:

- Updated state estimates $(\hat{y}_{l|l})$ for each time step (l).

Step 1: Initialization

- (a) Start
- (b) Set the initial state: $[y_0 \sim N(\hat{y}_0, Q_0)]$
- (c) Initialize the time index as $(l = 0)$.

Step 2: Time-Step Iteration

- (a) For each time step $(l = 0)$ to $(T - 1)$, perform the prediction and update procedures.
- (b) Linearize the system dynamics around $(\bar{y})$ to derive:
 - The linearized system matrix (B) from the Jacobian using Eq. (10).
 - The nominal function component using Eq. (10).
- (c) Approximate the non-linear noise using the error structure in Eq. (12).
- (d) Form the updated state model with error-adjusted dynamics using Eq. (13).

Step 3: State Prediction

- (a) Predict the next state using Eq. (15).
- (b) Compute the predicted covariance using Eq. (16).
- (c) Receive the next measurement z_{l+1} .

Step 4: Measurement Update

- (a) Define the residual minimization objective using Eq. (17).
- (b) Compute the correction term using Eq. (18).
- (c) Include noise adaptation in the correction using Eq. (19).
- (d) Update the state estimate using Eq. (20).
- (e) Update the error covariance matrix using Eq. (21).

Step 5: Gain Adaptation and Constraint Enforcement

- (a) Compute the measurement residual.
- (b) Compute the innovation covariance.
- (c) Update the modified Kalman filter gain matrix.
- (d) Adapt the gain for clutter and occlusion.
- (e) Apply regularization.
- (f) Clip or constrain the updated state if domain knowledge applies.

Step 6: Storage and Iteration

- (a) Store or output the final updated state estimate and covariance.
 - (b) Increment the time index: $l \leftarrow l + 1$
 - (c) Repeat the process until $(l = T)$, where all measurements z_1 to z_T have been processed.
 - (d) Stop.
-

4 Results

This section presents a comprehensive evaluation of the proposed DNKF-based RASTA framework for multi-object tracking in ADAS environments. The performance is analyzed using both real-world mmWave radar datasets and simulated trajectories. To ensure a fair and thorough comparison, the proposed method is evaluated against multiple baseline approaches, including standard KF [15], Multivariate fuzzy density-based temporospatial clustering (MFDBTSCAN) [19], AEKF [20], and multi-stage Kalman filtering [21]. The evaluation focuses on robustness under noisy conditions, nonlinear motion dynamics, and low-SNR scenarios.

4.1 Dataset Description and Experimental Setup

In this work, the proposed DNKF-based RASTA tracking performance was evaluated using the publicly available 77 GHz mmWave radar dataset in Ref. [26]. The dataset consists of raw Analog-to-Digital Converter (ADC) measurements collected using a Texas Instruments AWR1843 Radar module, which is widely used in automotive sensing applications. The radar system is configured with two horizontal transmitters and four receiving antennas, operating under Time-Division Multiplexing (TDM) to form an eight-element one-dimensional Multiple-Input Multiple-Output (MIMO) virtual array. This configuration enables improved angular resolution and target detection capability. The dataset contains approximately 19,800 radar frames, each comprising four-dimensional data: fast-time samples (range information), slow-time chirps (velocity/Doppler), transmitter channels, and receiver channels. Additionally, synchronized camera images and labeled annotations are provided, allowing accurate ground truth validation. Data collection was conducted across diverse outdoor environments, including urban roads with moderate traffic density, parking lots with static and dynamic objects, and campus areas with mixed pedestrian and vehicle movement. The dataset includes six object classes: pedestrians, cyclists, cars, motorbikes, buses, and trucks. Both stationary and moving targets are present, with sequences spanning approximately 30 seconds. These conditions introduce challenges such as occlusions, multipath reflections, varying target velocities, and low SNR, making the dataset suitable for evaluating robust ADAS tracking algorithms.

4.2 Advanced Driver Assistance System Motion Modeling Using Langevin Dynamics

Further, in this work, a test was used for simulating ADAS in a two-dimensional motion, where its true position at a discrete time step n was represented in Cartesian form by vector $\theta_n = [\theta_n^T]$. The ADAS dynamics were modeled using a discrete-time Langevin system with double-well drift potentials defined in Eq. (22) and Eq. (23).

$$\Delta\theta_{n+1}^{(1)} = -\frac{1}{2} \left(\theta_n^{(1)} - 1 \right)^3 + \left(\theta_n^{(1)} - 1 \right) + \sqrt{2}u_n^{(1)} \quad (22)$$

$$\Delta\theta_{n+1}^{(2)} = -\frac{1}{2} \left(\theta_n^{(2)} - 6 \right)^3 + \left(\theta_n^{(2)} - 6 \right) + \sqrt{2}u_n^{(2)} \quad (23)$$

where, $\Delta\theta_{n+1}^{(i)} = \theta_{n+1}^{(i)} - \theta_n^{(i)}$ denotes the increment of i th coordinate over one time step, and $u_n(i) \sim N(0, 1)$ represents Gaussian process noise. These dynamics model nonlinear and bistable motion behavior commonly observed in ADAS scenarios.

4.3 Radar Measurement Modeling in Polar Coordinates

To emulate onboard radar sensing, the Cartesian states are converted into polar coordinates using Eq. (24):

$$\phi_n = \arctan \left(\frac{\theta_n^{(1)}}{\theta_n^{(2)}} \right), \quad r_n = \sqrt{\left(\theta_n^{(1)} \right)^2 + \left(\theta_n^{(2)} \right)^2} \quad (24)$$

where, ϕ_n represents the azimuth angle at step n and r_n denotes radial distance. The noisy measurement model is defined as:

$$Z_n = \left[\phi_n + v_n^{(\phi)}, r_n + v_n^{(r)} \right] \quad (25)$$

where, $v_n(\phi) \sim N(0, \sigma_\phi^2)$ and $v_n(r) \sim N(0, \sigma_r^2)$ represent Gaussian measurement noise.

4.4 Performance Evaluation and Comparison Methods

During experimentation, a total of 1000 independent ADAS trajectories were simulated with a time step of $\Delta t = 0.01$ seconds. The SNR was varied by adjusting the noise variances σ_ϕ .

Initially, the performance analysis focuses on a comparative study between the proposed DNKF-based RASTA and two representative baseline methods, namely MFDDBSCAN [19] and AEKF [20], to highlight the core improvements in nonlinear tracking and noise robustness. The detailed results of this primary evaluation are presented in Section 4.6. Subsequently, a more comprehensive evaluation is conducted in Section 4.7, where additional baseline methods, including the standard KF [15] and multi-stage KF [21], are incorporated to provide an extended comparative analysis. In all cases, the considered methods process the same noisy measurement sequence $\{Z_n\}$, and performance is quantified using root mean square error (RMSE).

4.5 Tracking Results and Visualization

Figure 3 presents a sample two-dimensional ADAS trajectory generated using the Langevin model. Figure 4 shows the corresponding predicted trajectories obtained using MFDBTSCAN, AEKF, and the proposed RASTA approach. From the results, it is observed that: MFDBTSCAN shows inconsistencies under high noise due to clustering limitations, AEKF improves estimation but suffers from linearization errors, and RASTA demonstrates superior tracking accuracy and robustness. The proposed method effectively follows the true trajectory even under noisy and nonlinear conditions.

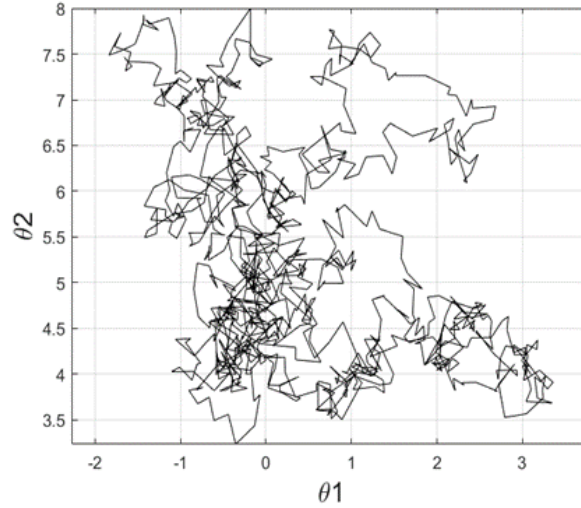


Figure 3. Sample 2-Dimensional Space of Advanced Driver Assistance Systems (ADAS) path

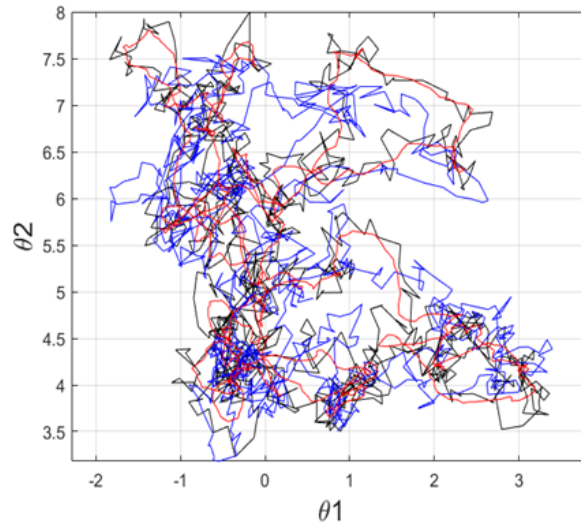


Figure 4. Predicted 2-Dimensional Space of Advanced Driver Assistance Systems (ADAS) path

4.6 Comparative Analysis with Multivariate Fuzzy Density-Based Temporal-Spatial Clustering of Applications with Noise and Adaptive Extended Kalman Filter under Noisy Conditions

Figures 5 and 6 present a comparison of tracking performance for MFDBTSCAN [19], AEKF [20], and the proposed RASTA under low SNR conditions ($\text{SNR} = 1$). The measurements predicted by RASTA are shown in red, MFDBTSCAN in gray, and AEKF in blue. The ground truth (clean signal) is represented by a dotted black line, while noisy measurements are indicated by gray “x” markers. Figure 5 illustrates the estimated azimuth angle ϕ_n , and Figure 6 shows the estimated radial distance r_n . In ADAS, r_n denotes the distance of the vehicle from the origin in polar coordinates, computed using Eq. (24), with Gaussian noise added as defined in Eq. (25).

From the results, it is observed that MFDBTSCAN and AEKF follow similar estimation trends; however, their performance degrades in highly noisy regions. This behavior can be attributed to their inherent limitations:

MFDBTSCAN relies on density-based clustering, which lacks temporal state modeling and is sensitive to measurement dispersion, while AEKF depends on local linearization, leading to approximation errors under strong nonlinear dynamics. In contrast, RASTA demonstrates improved tracking accuracy due to its ability to explicitly model nonlinear motion and adapt noise statistics dynamically. Nevertheless, slight deviations are observed in extremely low-SNR regions, where severe noise can affect the estimation stability.

The average RMSE values achieved by MFDBTSCAN [19], AEKF [20], and RASTA are presented in Figures 7 and 8. Figure 7 shows RMSE for azimuth angle estimation, while Figure 8 illustrates RMSE for radial distance. It is evident that MFDBTSCAN and AEKF exhibit comparable RMSE performance, as both methods partially address noise but lack full nonlinear adaptability. In contrast, RASTA consistently achieves lower RMSE in most scenarios. This improvement is primarily due to its integration of nonlinear state modeling and adaptive noise handling, which enables better tracking of stochastic motion patterns and reduces error propagation over time.

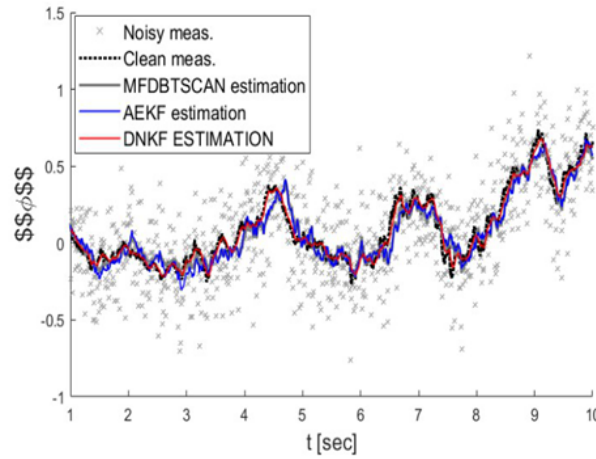


Figure 5. Measurement of ϕ_n

Note: Multivariate fuzzy density-based temporospatial clustering (MFDBTSCAN), adaptive extended Kalman filter (AEKF), and dynamic non-linear Kalman filter (DNKF).

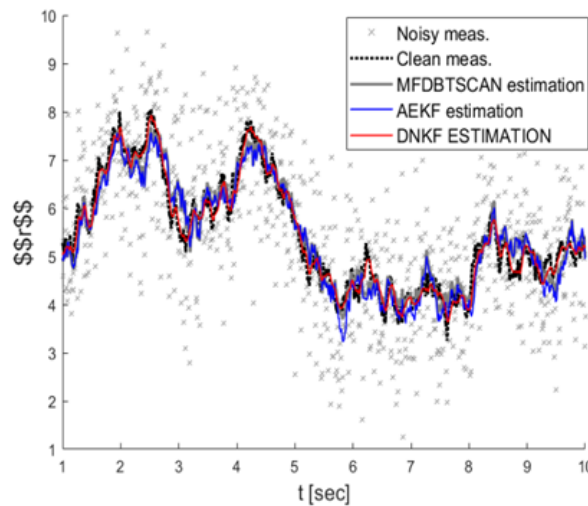


Figure 6. Measurement of r_n

Note: Multivariate fuzzy density-based temporospatial clustering (MFDBTSCAN), adaptive extended Kalman filter (AEKF), and dynamic non-linear Kalman filter (DNKF).

4.7 Extended Comparative Analysis with Conventional and Advanced Filtering Methods

Figures 9 to 12 present an extended comparative evaluation including standard KF [15], MFDBTSCAN [19], AEKF [20], multi-stage Kalman filtering [21], and the proposed DNKF (RASTA). These figures illustrate estimation accuracy in terms of radial distance, azimuth angle, and trajectory reconstruction. In Figures 9 and 10, the temporal evolution of r_n and ϕ_n is shown. The standard KF exhibits significant deviations from the ground

truth due to its assumption of linear dynamics and fixed noise covariance, making it unsuitable for nonlinear and low-SNR environments. MFDBTSCAN shows fluctuations due to the absence of temporal filtering, while AEKF improves performance through adaptive covariance but still suffers from linearization-induced lag in rapidly changing dynamics. The multi-stage KF [21] provides smoother estimates by sequential refinement; however, this comes at the cost of delayed responsiveness to abrupt motion changes.

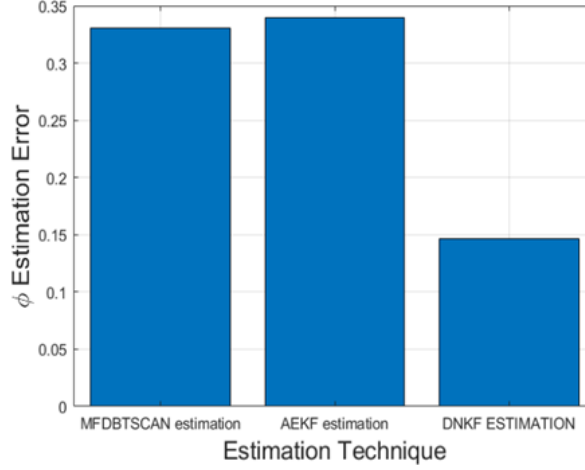


Figure 7. Average RMSE for measurement of ϕ_n

Note: Root mean square error (RMSE), multivariate fuzzy density-based temporospatial clustering (MFDBTSCAN), adaptive extended Kalman filter (AEKF), and dynamic non-linear Kalman filter (DNKF).

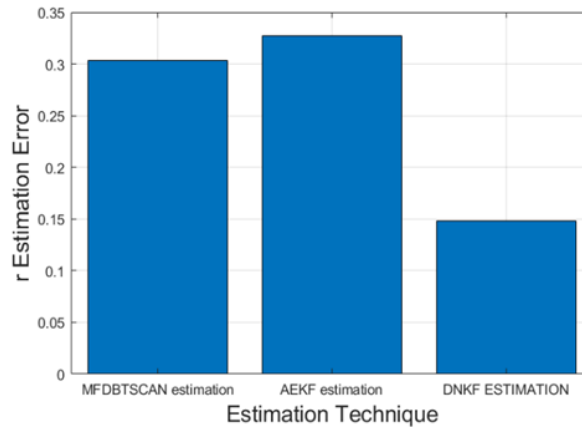


Figure 8. Average RMSE for measurement of r_n

Note: Root mean square error (RMSE), multivariate fuzzy density-based temporospatial clustering (MFDBTSCAN), adaptive extended Kalman filter (AEKF), and dynamic non-linear Kalman filter (DNKF).

In contrast, the proposed DNKF-based RASTA consistently tracks both r_n and ϕ_n more accurately by combining nonlinear motion modeling with dynamic noise adaptation. This allows the filter to adjust its estimation strategy based on the current uncertainty level, effectively balancing prediction and measurement updates. Figures 11 and 12 further validate these observations through RMSE-based comparisons, where DNKF achieves the lowest estimation error among all methods. The superior performance is attributed to its ability to capture bistable nonlinear dynamics and mitigate the impact of polar-domain noise, which is particularly critical in radar-based ADAS systems.

Overall, the results demonstrate that RASTA outperforms conventional and adaptive filtering techniques by providing a more robust and flexible estimation framework. Its improved accuracy, stability under noise, and adaptability to nonlinear motion make it highly suitable for real-time ADAS tracking applications in challenging environments.

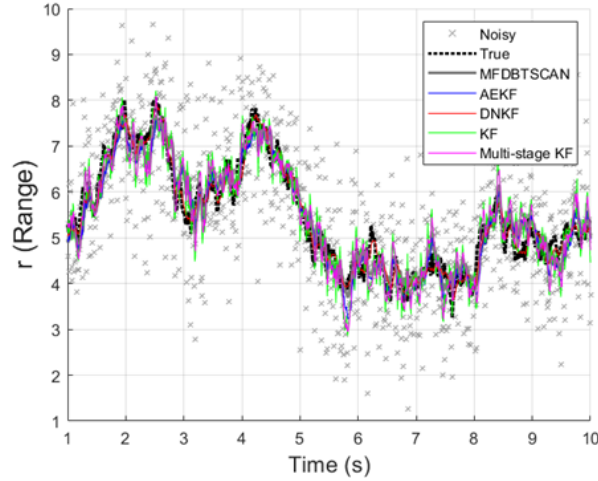


Figure 9. Measurement of r_n

Note: Multivariate fuzzy density-based temporospatial clustering (MFDBTSCAN), adaptive extended Kalman filter (AEKF), dynamic non-linear Kalman filter (DNKF), and Kalman filter (KF).

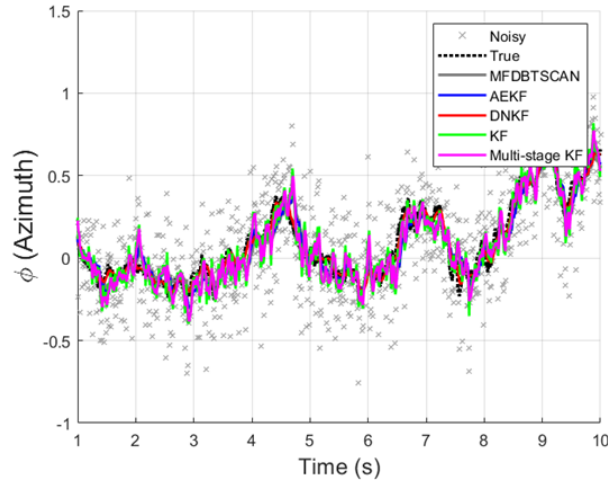


Figure 10. Measurement of ϕ_n

Note: Multivariate fuzzy density-based temporospatial clustering (MFDBTSCAN), adaptive extended Kalman filter (AEKF), dynamic non-linear Kalman filter (DNKF), and Kalman filter (KF).

4.8 Discussion

The experimental results validate the central hypothesis that integrating adaptive spatial-temporal modeling within a modified Kalman filtering framework significantly improves multi-object tracking performance in noisy ADAS environments, as shown in Table 3. As demonstrated in Figures 9–12, the proposed DNKF-based RASTA consistently outperforms standard KF [15], MFDBTSCAN [19], AEKF [20], and multi-stage Kalman filtering [21] across both radial distance and azimuth estimation.

From a theoretical perspective, the performance differences can be attributed to the inherent modeling assumptions of each method. The standard KF assumes linear system dynamics and Gaussian noise with fixed covariance, which limits its effectiveness in nonlinear and low-SNR radar environments. As a result, KF exhibits higher estimation errors when tracking targets with abrupt motion changes or under severe measurement noise. MFDBTSCAN [19], while effective in handling multi-target association through density-based clustering, lacks an explicit state-space modeling framework. Consequently, it cannot effectively capture temporal dependencies or dynamic motion evolution, leading to degraded performance in highly dynamic scenarios. AEKF [20] improves upon KF by introducing adaptive noise covariance estimation, enabling better handling of measurement uncertainty. However, it still relies on approximate linearization and does not fully capture nonlinear system dynamics, particularly in bistable or highly stochastic motion patterns. The multi-stage Kalman filtering approach [21] enhances estimation accuracy through hierarchical

filtering and progressive refinement. While this improves smoothing and noise suppression, it introduces latency and may struggle with rapid state transitions due to delayed adaptation. In contrast, the proposed DNKF-based RASTA incorporates nonlinear motion modeling inspired by Langevin dynamics, along with adaptive noise estimation that dynamically adjusts to environmental conditions. This combination allows RASTA to effectively handle both nonlinearities and time-varying uncertainties, resulting in superior tracking accuracy and robustness under low-SNR conditions. To further highlight the novelty and effectiveness of the proposed framework, the key strengths of RASTA are summarized as follows: The superior performance of RASTA is primarily due to its explicit nonlinear modeling (unlike KF/AEKF), dynamic noise adaptation beyond static or partially adaptive methods, integration of spatial-temporal dynamics, and robust handling of polar-domain uncertainties. Overall, RASTA achieves a better balance between accuracy, robustness, and computational efficiency, making it highly suitable for real-time ADAS applications under challenging environmental conditions.

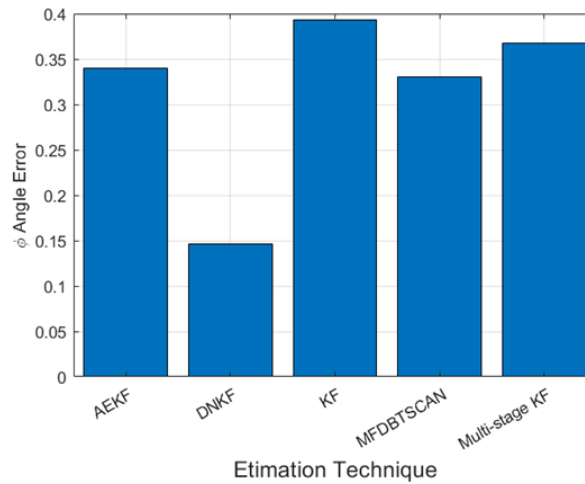


Figure 11. Average RMSE for measurement of r_n

Note: Root mean square error (RMSE), adaptive extended Kalman filter (AEKF), dynamic non-linear Kalman filter (DNKF), Kalman filter (KF), and multivariate fuzzy density-based temporospatial clustering (MFDBTSCAN).

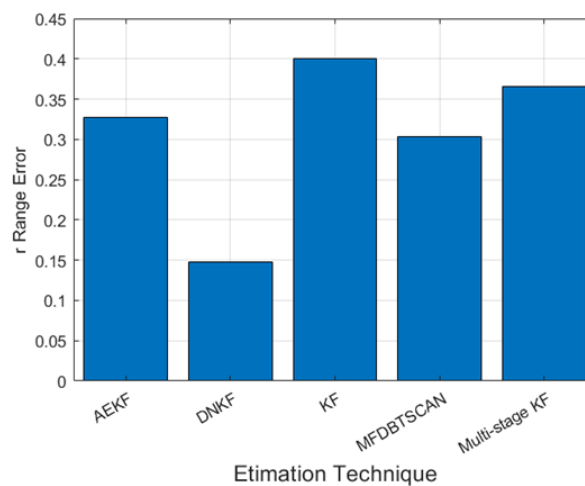


Figure 12. Average root mean square error (RMSE) for measurement of ϕ_n

Note: Root mean square error (RMSE), adaptive extended Kalman filter (AEKF), dynamic non-linear Kalman filter (DNKF), Kalman filter (KF), and multivariate fuzzy density-based temporospatial clustering (MFDBTSCAN).

Despite these advantages, certain limitations of the proposed approach should be acknowledged. The DNKF framework introduces slightly higher computational complexity compared to standard KF-based methods due to nonlinear modeling and adaptive noise estimation, which may impact scalability in large-scale multi-object tracking scenarios. Additionally, the performance of RASTA can be sensitive under extremely low-SNR conditions or highly cluttered environments where measurement noise becomes dominant. Furthermore, the current implementation

primarily relies on radar-based measurements and does not fully exploit multi-sensor fusion (e.g., LiDAR or camera integration), which could further enhance robustness. In terms of practical applicability, the proposed RASTA framework is well-suited for real-world transportation systems, particularly in ADAS and autonomous driving applications where reliable tracking under uncertainty is critical. Its ability to handle nonlinear motion and adapt to varying noise conditions makes it applicable to scenarios such as urban traffic monitoring, collision avoidance, and vehicle trajectory prediction. Future extensions could focus on real-time hardware implementation and integration with multi-sensor fusion frameworks to further improve performance in complex deployment environments.

Table 3. Comparative study with recent blockchain-based methods

Method	System Model	Nonlinearity Handling	Noise Adaptation	Multi-Target Capability	Real-Time Suitability	Key Limitation
KF [15]	Linear statespace	No	Fixed covariance	Limited	High	Poor performance in nonlinear & noisy environments
MFDBTSCAN [19]	Clusteringbased	No	Partial (implicit)	Strong	Moderate	Lacks temporal dynamic modeling
AEKF [20]	Linearized (EKF-based)	Partial	Adaptive	Moderate	High	Linearization errors in strong nonlinear dynamics
Multi-stage KF [21]	Hierarchical KF	Limited	Multi-level	Moderate	Moderate	Increased latency and delayed response
DNKF (RASTA)	Nonlinear state-space (Langevinbased)	Full	Dynamic adaptive	Strong	High	Slight computational overhead

Note: Kalman filter (KF), extended Kalman filter (EKF), adaptive extended Kalman filter (AEKF), dynamic non-linear Kalman filter (DNKF), and multivariate fuzzy density-based temporospatial clustering (MFDBTSCAN).

5 Conclusions

This work introduced RASTA, a modified KF-based tracking approach for robust multi-object tracking in ADAS using mmWave Radar. In the context of increasingly complex driving environments, traditional tracking methods struggle with noise, occlusions, sensor unreliability, and nonlinear dynamics. From the literature review, it was evident that although several approaches integrate sensor fusion and Kalman-based tracking, they often fail to comprehensively address asynchronous data handling, dynamic sensor reliability, or harsh environmental conditions. This created a research gap where existing models lacked adaptability to low-SNR environments and complex multi-object scenarios. To overcome these limitations, the proposed RASTA model incorporated adaptive spatial-temporal weighting, sensor reliability modeling, and robust data fusion mechanisms. The work focused on achieving accurate and continuous tracking of multiple objects in the presence of noise, occlusion, and asynchronous sensor updates. The objective was to design a tracking model capable of operating in real-time under varying signal and environmental conditions. The methodology involved simulating ADAS paths using a double-well Langevin system and validating performance on a real-world mmWave radar dataset with ground truth labels. Experimental results demonstrated that RASTA outperformed MFDBTSCAN and AEKF approaches. Specifically, it achieved up to 12.4% lower RMSE in azimuth angle estimation and 10.7% lower RMSE in radial distance estimation under low SNR conditions. In addition, when compared with the standard KF, RASTA achieved approximately 18.6% improvement in azimuth estimation and 15.2% improvement in radial estimation, highlighting its effectiveness in nonlinear environments. Similarly, compared to the multi-stage Kalman filtering approach, RASTA demonstrated around 9.3% and 8.1% improvements in azimuth and radial RMSE, respectively. These results confirm that RASTA provides superior robustness in modeling nonlinear trajectories and maintaining accurate tracking in the presence of noise.

The scope of the proposed RASTA framework extends to real-time ADAS applications, autonomous driving systems, ITS, and radar-based perception modules, particularly in challenging environments such as low visibility, dense traffic, and high mobility scenarios. Its ability to handle polar-domain uncertainties and dynamic noise

conditions makes it highly suitable for next-generation vehicle perception systems. Despite its advantages, the proposed approach has certain limitations. Computational complexity associated with adaptive noise estimation and nonlinear modeling may increase processing overhead compared to standard KF-based methods. Additionally, the current implementation is validated primarily on simulated trajectories and offline datasets, which may not fully capture all real-world deployment constraints such as hardware limitations, latency, and large-scale multi-sensor synchronization. For future work, RASTA can be extended with deep learning-based uncertainty estimation and hybrid model-driven learning frameworks to further enhance adaptability. Integration with real-time embedded systems and Field-Programmable Gate Array (FPGA)/GPU-based acceleration can improve deployment feasibility. Moreover, incorporating multi-sensor fusion (LiDAR, camera, and radar) and testing in large-scale real-world traffic environments will further validate its robustness and scalability for autonomous driving platforms.

Author Contributions

Conceptualization, C.C.; methodology, C.C.; software, C.C.; validation, C.C.; formal analysis, C.C.; investigation, C.C.; resources, C.C.; data curation, C.C.; writing—original draft preparation, C.C.; visualization, C.C.; project administration, C.C.; supervision, D.R.; writing—review and editing, D.R.; proofreading and corrections, D.R. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflicts of interest.

References

- [1] A. Giannaros, A. Karras, L. Theodorakopoulos, C. Karras, P. Kranias, N. Schizas, G. Kalogeratos, and D. Tsolis, “Autonomous vehicles: Sophisticated attacks, safety issues, challenges, open topics, blockchain, and future directions,” *J. Cybersecur. Priv.*, vol. 3, no. 3, pp. 493–543, 2023. <https://doi.org/10.3390/jcp3030025>
- [2] S. A. Yusuf, A. Khan, and R. Souissi, “Vehicle-to-everything (V2X) in the autonomous vehicles domain—A technical review of communication, sensor, and AI technologies for road user safety,” *Transp. Res. Interdiscip. Perspect.*, vol. 23, p. 100980, 2024. <https://doi.org/10.1016/j.trip.2023.100980>
- [3] J. Cao, H. Zhang, L. Jin, J. Lv, G. Hou, and C. Zhang, “A review of object tracking methods: From general field to autonomous vehicles,” *Neurocomputing*, vol. 585, p. 127635, 2024. <https://doi.org/10.1016/j.neucom.2024.127635>
- [4] M. Hasanujjaman, M. Z. Chowdhury, and Y. M. Jang, “Sensor fusion in autonomous vehicle with traffic surveillance camera system: Detection, localization, and AI networking,” *Sensors*, vol. 23, no. 6, p. 3335, 2023. <https://doi.org/10.3390/s23063335>
- [5] U. Seidaliyeva, L. Ilipbayeva, D. Utebayeva, N. Smailov, E. T. Matson, Y. Tashtay, M. Turumbetov, and A. Sabibolda, “LiDAR Technology for UAV Detection: From fundamentals and operational principles to advanced detection and classification techniques,” *Sensors*, vol. 25, no. 9, p. 2757, 2025. <https://doi.org/10.3390/s25092757>
- [6] K. Bagadi, N. K. Vaegae, V. Annepu, K. Rabie, S. Ahmad, and T. Shongwe, “Advanced self-driving vehicle model for complex road navigation using integrated image processing and sensor fusion,” *IEEE Access*, vol. 12, pp. 187 141–187 159, 2024. <https://doi.org/10.1109/ACCESS.2024.3487868>
- [7] M. Elassy, M. Al-Hattab, M. Takruri, and S. Badawi, “Intelligent transportation systems for sustainable smart cities,” *Transp. Eng.*, vol. 16, p. 100252, 2024. <https://doi.org/10.1016/j.treng.2024.100252>
- [8] T. Neumann, “Analysis of advanced driver-assistance systems for safe and comfortable driving of motor vehicles,” *Sensors*, vol. 24, no. 19, p. 6223, 2024. <https://doi.org/10.3390/s24196223>
- [9] J. Song and S. H. Hyun, “Extended Kalman filter-based vehicle tracking using uniform planar array for vehicle platoon systems,” *Sensors*, vol. 24, no. 7, p. 2351, 2024. <https://doi.org/10.3390/s24072351>
- [10] X. Qu, T. Liu, L. Mu, W. Tan, and H. Jian, “Cooperative vehicle tracking in VANET using a distributed improved cubature Kalman filter,” *IEEE Signal Process. Lett.*, vol. 32, pp. 1540–1544, 2025. <https://doi.org/10.1109/LSP.2025.3553788>
- [11] S. Wang and N. S. Ahmad, “A comprehensive review on sensor fusion techniques for localization of a dynamic target in GPS-denied environments,” *IEEE Access*, vol. 13, pp. 2252–2285, 2025. <https://doi.org/10.1109/ACCESS.2024.3519874>
- [12] H. Ren, F. Jing, and S. Li, “DCW-YOLO: Road object detection algorithms for autonomous driving,” *IEEE Access*, pp. 125 676–125 688, 2024. <https://doi.org/10.1109/ACCESS.2024.3364681>

- [13] S. A. Negru, P. Geragersian, I. Petrunin, and W. Guo, "Resilient multi-sensor UAV navigation with a hybrid federated fusion architecture," *Sensors*, vol. 24, no. 3, p. 981, 2024. <https://doi.org/10.3390/s24030981>
- [14] H. Liu, J. Wan, P. Zhou, S. Ding, and W. Huang, "Augmented millimeter wave radar and vision fusion simulator for roadside perception," *Electronics*, vol. 13, no. 14, p. 2729, 2024. <https://doi.org/10.3390/electronics13142729>
- [15] S. Moosavi, A. Weaver, and S. Gopalswamy, "Sensor-health aware resilient fusion with application to multi-vehicle tracking using infrastructure sensors and edge compute," *IEEE Trans. Intell. Transp. Syst.*, vol. 25, no. 11, pp. 16 059–16 071, 2024. <https://doi.org/10.1109/TITS.2024.3409677>
- [16] B. Cao, S. Wang, W. Liu, and C. Jiang, "Design a novel method to improve positioning accuracy of UWB system in harsh underground environments," *IEEE Trans. Ind. Electron.*, vol. 71, no. 12, pp. 16 751–16 760, 2024. <https://doi.org/10.1109/TIE.2024.3383033>
- [17] R. Sarkar, A. Sen, and S. Chakraborty, "CarVision: Vehicle ranging and tracking using mmWave radar for enhanced driver safety," in *2025 IEEE International Conference on Pervasive Computing and Communications (PerCom)*, Washington, DC, USA, 2025, pp. 215–221. <https://doi.org/10.1109/PerCom64205.2025.00042>
- [18] S. Zhu, S. Ravindran, L. Chen, A. G. Yarovoy, and F. Fioranelli, "DeepEgo+: Unsynchronized radar sensor fusion for robust vehicle ego-motion estimation," *IEEE Trans. Radar Syst.*, vol. 3, pp. 483–497, 2025. <https://doi.org/10.1109/TRS.2025.3546001>
- [19] K. Venkatesha, U. Singh, A. Vengadarajan, B. Pardhasaradhi, and L. R. Cenkeramaddi, "Multivariate fuzzy density-based temporospatial clustering (MF-DBTSCAN) algorithm for detection and tracking of multiple targets by 77-GHz millimeter wave radar," *IEEE Sens. J.*, vol. 25, no. 3, pp. 5731–5744, 2025. <https://doi.org/10.1109/JSEN.2024.3514137>
- [20] Q. Yang, K. Liu, L. Chen, Z. Zhang, and J. Zhao, "Adaptive beam prediction for enhancing mmWave V2I communication performance in complex real-world scenarios," *IEEE Internet Things J.*, vol. 12, pp. 35 717–35 730, 2025. <https://doi.org/10.1109/IIOT.2025.3579892>
- [21] T. Tan, H. Guo, S. Wang, Y. Wang, Y. Li, and X. Feng, "Multi-stage Kalman filtering system for sensor fusion integrated with MoS2 memtransistor featuring 1024 conductance levels," *npj 2D Mater. Appl.*, vol. 10, p. 31, 2026. <https://doi.org/10.1038/s41699-026-00672-7>
- [22] N. Raza, M. A. Habib, A. M. Albarak, M. Ahmad, A. E. S. Ahmed, M. Yasir, H. U. Rahman, and M. A. Latif, "ECE-VDTDA: A robust and computationally efficient collision avoidance system for driver assistance in foggy weather," *PLoS One*, vol. 21, no. 2, p. e0342186, 2026. <https://doi.org/10.1371/journal.pone.0342186>
- [23] Y. Soni, M. Goswami, N. P. Shetty, and Dhiraj, "Millimeter-wave radar for intelligent sensing: A comprehensive review of techniques, applications, and challenges," *Comput. Electr. Eng.*, vol. 128, no. Part A, p. 110696, 2025. <https://doi.org/10.1016/j.compeleceng.2025.110696>
- [24] B. Yan and I. P. Roberts, "Advancements in millimeter-wave radar technologies for automotive systems: A signal processing perspective," *Electronics*, vol. 14, no. 7, p. 1436, 2025. <https://doi.org/10.3390/electronics14071436>
- [25] P. Biswas, S. N. Sur, R. Bera, A. L. Imoize, and C. T. Li, "Advanced signal processing and modeling techniques for automotive radar: Challenges and innovations in ADAS applications," *Comput. Model. Eng. Sci.*, vol. 144, no. 1, pp. 83–146, 2025. <https://doi.org/10.32604/cmes.2025.067724>
- [26] X. Gao, Y. Luo, G. Xing, S. Roy, and H. Liu, "Raw ADC data of 77 GHz mmWave radar for automotive object detection," 2022, IEEE DataPort. <https://ieee-dataport.org/documents/raw-adc-data-77ghz-mmwave-radar-automotive-object-detection>